Samanyu Okade

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EDUCATION

Delft University of Technology
Delft, Netherlands 2023-2025
MSc in Electrical Engineering – Microelectronics track, specialization in Digital Systems

Vellore Institute of Technology
Vellore, India 2019-2023
BTech in Electronics and Communication Engineering: 8.25CGPA

TOP TECHNICAL SKILLS

Python; Verilog HDL; Digital CMOS Design; MATLAB; PCB designing; Sensor Technology

WORK EXPERIENCE AND INTERNSHIPS

RTips Technologies
Hardware intern ***Oct 2022- May 2023***

- Researched, and improved on a device to convert between **MODBUS and DALI** communication protocols from base **design of the PCB**, to its integration.

Students for the Exploration and Development of Space (SEDS India)
Executive Director ***Nov 2021- Nov 2022***

- Successfully led and organized a cross-functional team of 8 SEDS India Staff members to **improve efficiency and communication**.
- Responsible for **maintaining all projects and relations** within SEDS India and its **14 sub-chapters**.

Chapter Affairs and Expansions Manager ***Aug 2020- Oct 2021***

- Oversaw the **expansion of SEDS as a chapter to 2 more university teams to join our community** of students for Space Development.
- **Ideating for projects** like CubeSats, **and competitions** like International Rover Competition and International Rover Design Competition to help all the 14 sub-chapters that required it.

The Institution of Engineering and Technology (IET-VIT Vellore)
Hardware Head ***Dec 2020- Dec 2021***

- **Managed, taught, and guided a team of 12** in learning hardware skills in **embedded systems, fundamentals of communication, CMOS, and digital designs**.
- **Mentored 3 teams to victory** in hardware tracks and open categories **in hackathons** Equinox, and Hack4cause.

ACADEMIC PROJECTS

A Study on the Error Tolerance of Spiking versus Non-Spiking Neural Networks ***Oct 2024 – Dec 2024***

- An initial software-based **comparative stud**y on the advantages of using **Spiking Neural Network**-based Neuromorphic Computers to **Artificial Neural Networks**.
- Compared the error tolerance results of a CNN's predictions on an MNIST dataset with a Spiking CNN on an NMNIST dataset to **prove the error-tolerant abilities of the SNN**.

FPGA based (hardware) decision making for efficient satellite orientation and propulsion ***Oct 2022 – Apr 2023***
(published in IEEE Xplore in June 2023)

- Devised a method for **rockets and propulsion engines** to orient and **align** the solar panels and themselves using **FPGAs** to make the **most efficient positioning decisions** using sensor-based information.

The RTA [Robotic Temperate Analyzer] ***Mar 2022 – May 2022***

- Created a prototype of a small **robot that moves along queues in crowded indoor areas** to ensure a personal space between two people, while also **reading surrounding internal temperature and humidity conditions** for hygienic reasons including but not limited to COVID-19.

A Robotic Solution for Internal Imperfection Detection in Industrial Machinery ***Sept 2021 – Nov 2021***
(Published in the AIP Conference Proceedings in March 2024)

- Prototyped a small, mobile robot with flame, DHT-11, MO-26, and ultrasonic sensors to **detect anomalies and conditions within tight industrial equipment spaces**.
- The distance sensor-mounted servo motor automates steering within tight industrial machine spaces.

Goniometry- Visual Aid ***Oct 2021***

- Extending the use of a **flex sensor incorporated glove** to operate a **feedback system for the visually impaired** while walking, **omitting the use of a walking stick**.
- Using a rotating servo motor mounted with a distance sensor atop a cap would require the user to merely move a finger in the glove.

OTHER TECHNICAL SKILLS

Embedded Systems; Assembly Level Language; Internet of Things (IoT); Robotics; Python Programming; basics of C/C++

CERTIFICATIONS

- Provisional discovery of the Main Belt asteroid 2021EM17.
- CMOS Digital VLSI Design-course certification offered by IIT Roorkee. (NPTEL)
- Linear Circuits 2:AC Analysis-course certification offered by Georgia Institute of Technology. (Coursera)
- Introduction to Electronics-course certification offered by Georgia Institute of Technology. (Coursera)
- Introduction and Programming with IoT Boards-course certification offered by POSTECH. (Coursera)
- National Science Olympiad (NSO) Zonal Gold medalist in 2015.
- Academic proficiency awards in 2013, 2015, and 2017.

CAMPUS & COMMUNITY INVOLVEMENT

- Core Committee member at Institution of Engineering and Technology (IET) 2019-2023
- Core Committee member of SEDS-VIT 2019-2023
- SEDS India Staff and Board member 2020-2023
- Alpha team-member of VIT Dance club 2020-2023
- Flautist and musician in VIT Community Radio 2019-2023
- VIT Swim team 2019-2023

KEY SKILLS

Leadership; Public-speaking; Effective Communication; Team-management; Writing; Creative thinking; Adaptability

EXTRA-CURRICULARS

- Participated in and won art and painting competitions, both national and International (Japan, India, and Malaysia).
- Poom Belt in Taekwondo.
- Participated and won competitions of various levels as a Flautist, keyboardist, Vocalist, and Dancer.